

Erik Löffelholz

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PROFILE

Software developer and computational scientist with a Master's degree in Mathematical Physics. Strong background in machine learning (PyTorch), numerical simulation, and full-stack web development. Experienced in building differentiable simulation systems, GPU-accelerated compute pipelines, and interactive 3D visualization tools. Combines deep mathematical foundations with hands-on engineering skills in Python, TypeScript, and modern web frameworks.

EDUCATION

- Oct 2021 – Sep 2025

M.Sc. Mathematical Physics
Universität Leipzig
Focus: Differential Geometry · PDE Theory · Quantum Field Theory · Numerical Methods
- Oct 2018 – Mar 2022

B.Sc. Physics
Universität Leipzig

PROFESSIONAL EXPERIENCE

- Apr 2025 – Present

Scientific Software Developer
Max Planck Institute for Mathematics in the Sciences, Leipzig
 - Built differentiable simulation software for mathematical research using Python and PyTorch
 - Developed interactive 3D visualization tools with Three.js and WebSocket streaming
 - Implemented numerical solvers for PDE systems with real-time browser-based frontends
 - Co-authored a research paper on hyperbolic surface mesh embeddings (Bridges Conference 2026, under review)
- Mar 2024 – Feb 2025

Working Student — Mathematics Editorial
Ernst Klett Verlag GmbH, Leipzig
 - Reviewed and quality-assured mathematical content for educational textbooks
 - Ensured technical correctness and pedagogical clarity across multiple publications

TECHNICAL SKILLS

Languages

Python · TypeScript · JavaScript · C++ · HTML/CSS

Web & Frontend

React · Three.js · WebGL/WebGPU · Vite · Tailwind CSS · WebSocket

Scientific Computing

NumPy · SciPy · SymPy · Numerical PDE solvers · Monte Carlo methods

Machine Learning

PyTorch · Graph Neural Networks · Diffusion Models · VAEs · Neural ODEs · PINNs

Backend & Infrastructure

FastAPI · REST APIs · Docker · Git · CI/CD

Spoken Languages

German (native) · English (C1 — full professional proficiency)

SELECTED PROJECTS

Graph ML Lab — From-scratch PyTorch framework for spatial graph generation: GCN, GAT, graph VAE, and joint discrete-continuous diffusion models over 3D graph structures. No external GNN libraries. github.com/erik2810/ml-projects

Mesh-Based Physics Simulator — Fully differentiable physics engine in PyTorch. Maps mesh topology to particle-spring systems with energy-based force computation and end-to-end backpropagation through dynamics.

PINN Solver — Physics-Informed Neural Network solver for 10 PDEs (Burgers, Heat, Wave, KdV, etc.) with PyTorch training backend and client-side JavaScript inference. github.com/erik2810/pde-solver

DiffQFT — Differentiable quantum field theory computations in AdS_2 . Neural surrogates replacing Monte Carlo integration, PINN solvers, FastAPI backend. github.com/erik2810/DiffQFT

Knitted Models — Computational geometry engine for generating woven strand patterns on quad meshes using half-edge data structures and spline interpolation. Three.js rendering with GPU path tracing.

Neural ODE Engine — Neural network learning chaotic Lorenz dynamics in real time. From-scratch backpropagation, RK4 integrator, and Adam optimizer in pure JavaScript. github.com/erik2810/differentiable-physics-engine

Full portfolio with live demos at erik2810.github.io/projects

REFERENCES

Available upon request.